$$
\begin{aligned}
&C^{(12)\,\mathrm{i1\_i2\_}}_{e\phi_1} \to \\
&\frac{1}{16\pi^2} \Big(-\frac{1}{9} g_2^4 y_e^{i1i2} LF_{3,0}[m_2] - \frac{1}{6} g_2^4 y_e^{i1i2} LF_{4,-1}[m_2] + \frac{8}{45} g_2^4 y_e^{i1i2} LF_{5,-2}[m_2] + \frac{1}{72} g_2^2 y_e^{i1i2} \\
&\quad (18 \overline{y_e}^{pr} y_e^{pr} - 13 \textstyle\sum_p g_2^2) LF_{3,0}[m_{\tilde{l}}^{p}] + \frac{1}{48} g_2^2 y_e^{i1i2} (-12 \overline{y_e}^{pr} y_e^{pr} + 11 \textstyle\sum_p g_2^2) LF_{4,-1}[m_{\tilde{l}}^{p}] - \\
&\frac{2}{45} \textstyle\sum_p g_2^4 y_e^{i1i2} LF_{5,-2}[m_{\tilde{l}}^{p}] + \frac{1}{24} g_2^2 y_e^{i1i2} (18 \overline{y_d}^{pr} y_d^{pr} + 18 \overline{y_u}^{pr} y_u^{pr} - 13 \textstyle\sum_p g_2^2) LF_{3,0}[m_{\tilde{q}}^{p}] + \\
&\frac{1}{16} g_2^2 y_e^{i1i2} (-12 (\overline{y_d}^{pr} y_d^{pr} + \overline{y_u}^{pr} y_u^{pr}) + 11 \textstyle\sum_p g_2^2) LF_{4,-1}[m_{\tilde{q}}^{p}] - \\
&\frac{2}{15} \textstyle\sum_p g_2^4 y_e^{i1i2} LF_{5,-2}[m_{\tilde{q}}^{p}] - \frac{1}{18} g_2^4 y_e^{i1i2} LF_{3,0}[\tilde{\mu}] - \frac{1}{12} g_2^4 y_e^{i1i2} LF_{4,-1}[\tilde{\mu}] + \\
&\frac{4}{45} g_2^4 y_e^{i1i2} LF_{5,-2}[\tilde{\mu}] + \frac{1}{16} g_1^2 g_2^2 y_e^{i1i2} LF_{2,2,-1}[m_1, m_{\tilde{l}}^{i1}] - \frac{1}{16} g_2^4 y_e^{i1i2} LF_{2,2,-1}[m_2, m_{\tilde{l}}^{i1}] + \\
&\frac{2}{3} g_2^4 y_e^{i1i2} LF_{2,1,0}[m_2, \tilde{\mu}] - \frac{1}{3} g_2^4 y_e^{i1i2} LF_{2,2,-1}[m_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 y_e^{i1i2} LF_{2,2,0}[m_2, \tilde{\mu}] - \\
&\frac{1}{3} g_2^4 y_e^{i1i2} LF_{3,1,-1}[m_2, \tilde{\mu}] + \frac{5}{3} g_2^4 y_e^{i1i2} LF_{3,2,-2}[m_2, \tilde{\mu}] - \frac{3}{2} g_2^4 m_2^2 y_e^{i1i2} LF_{3,2,-1}[m_2, \tilde{\mu}] - \\
&g_2^4 y_e^{i1i2} LF_{3,3,-3}[m_2, \tilde{\mu}] + g_2^4 m_2^2 y_e^{i1i2} LF_{3,3,-2}[m_2, \tilde{\mu}] - g_2^4 y_e^{i1i2} LF_{4,2,-3}[m_2, \tilde{\mu}] + \\
&g_2^4 m_2^2 y_e^{i1i2} LF_{4,2,-2}[m_2, \tilde{\mu}] + g_2^2 y_e^{i1i2} \overline{a_e}^{pr} a_e^{pr} LF_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \\
&\frac{1}{4} g_2^2 y_e^{i1i2} (9 \overline{a_e}^{pr} a_e^{pr} + \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pr}) LF_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \\
&\frac{1}{6} g_2^2 y_e^{i1i2} (7 \overline{a_e}^{pr} a_e^{pr} + \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pr}) LF_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \frac{1}{8} g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,0}[m_{\tilde{l}}^{i1}, m_1] + \\
&\frac{1}{8} g_2^4 y_e^{i1i2} LF_{2,1,0}[m_{\tilde{l}}^{i1}, m_2] + 3 g_2^2 y_e^{i1i2} \overline{a_d}^{pr} a_d^{pr} LF_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\frac{3}{4} g_2^2 y_e^{i1i2} (9 \overline{a_d}^{pr} a_d^{pr} + \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pr}) LF_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \\
&\frac{1}{2} g_2^2 y_e^{i1i2} (7 \overline{a_d}^{pr} a_d^{pr} + \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{pr}) LF_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\frac{3}{4} \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{2,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \frac{3}{4} \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{3,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \\
&\frac{1}{4} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} + 4 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} - \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \\
&\frac{1}{4} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} - 2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4} g_2^2 y_e^{i1i2} (-\overline{a_u}^{pr} a_u^{pr} + \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4} \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{2,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}] + \frac{3}{4} \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{3,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}] + \\
&\frac{1}{4} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} + 7 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{1}{4} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} + 4 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \\
&\frac{3}{4} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} + 7 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{1}{2} g_2^2 y_e^{i1i2} (\overline{a_u}^{pr} a_u^{pr} + 7 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr}) LF_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \\
&\frac{3}{2} \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{2,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] + \frac{3}{2} \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{3,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] - \\
&g_1^2 g_2^2 y_e^{i1i2} LF_{3,1,-1}[\tilde{\mu}, m_1] - \frac{1}{2} g_1^4 \tilde{\mu}^2 y_e^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_1] + \\
&\frac{29}{12} g_1^2 g_2^2 y_e^{i1i2} LF_{4,1,-2}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 \tilde{\mu}^2 y_e^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_1] - \\
&\frac{4}{3} g_1^2 g_2^2 y_e^{i1i2} LF_{5,1,-3}[\tilde{\mu}, m_1] - \frac{13}{3} g_2^4 y_e^{i1i2} LF_{3,1,-1}[\tilde{\mu}, m_2] + \frac{4}{3} g_2^4 y_e^{i1i2} LF_{3,2,-2}[\tilde{\mu}, m_2] - \\
&\frac{1}{2} g_2^4 y_e^{i1i2} (4 m_2^2 + 3 \tilde{\mu}^2) LF_{3,2,-1}[\tilde{\mu}, m_2] + \frac{33}{4} g_2^4 y_e^{i1i2} LF_{4,1,-2}[\tilde{\mu}, m_2] - \\
&g_2^4 y_e^{i1i2} LF_{4,2,-3}[\tilde{\mu}, m_2] + \frac{3}{2} g_2^4 y_e^{i1i2} (m_2^2 + \tilde{\mu}^2) LF_{4,2,-2}[\tilde{\mu}, m_2] - 4 g_2^4 y_e^{i1i2} LF_{5,1,-3}[\tilde{\mu}, m_2] - \\
&\frac{1}{8} m_1 g_1^2 g_2^2 a_e^{i1i2} LF_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] - \frac{1}{8} m_1 g_1^2 g_2^2 a_e^{i1i2} LF_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{i2}] + \\
&\frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{r}] + \frac{1}{4} g_1^2 g_2^2 y_e^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] + \\
&g_1^4 \tilde{\mu}^2 y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] - \frac{1}{8} g_1^2 g_2^2 y_e^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \\
&\frac{1}{2} g_1^4 \tilde{\mu}^2 y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{4} g_2^4 y_e^{i1i2} LF_{2,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{2} g_2^4 m_2^2 y_e^{i1i2} LF_{2,1,1,0}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{2} g_2^4 y_e^{i1i2} LF_{3,1,1,-2}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{2} g_2^4 m_2^2 y_e^{i1i2} LF_{3,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - 2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] - \\
&\frac{3}{2} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + g_1^2 g_2^2 y_e^{i1i2} LF_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] + \\
&m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] - \frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{e}}^{r}] - \\
&\frac{7}{8} g_2^4 y_e^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{1}{4} g_2^4 y_e^{i1i2} (5 m_2^2 + 6 \tilde{\mu}^2) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{4} m_1 g_1^2 g_2^2 a_e^{i1i2} LF_{2,1,1,0}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] - \frac{1}{4} m_1 g_1^2 g_2^2 a_e^{i1i2} LF_{3,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{4} \tilde{\mu}^2 \overline{y_e}^{pr} \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_e^{i1i2} LF_{2,2,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{e}}^{r}] + \\
&\frac{3}{4} m_1 g_1^2 g_2^2 a_e^{i1i2} LF_{2,1,1,0}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] - \frac{1}{4} m_1 g_1^2 g_2^2 a_e^{i1i2} LF_{3,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] - \\
&\frac{1}{4} g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, \tilde{\mu}] - \frac{1}{4} g_2^4 y_e^{i1i2} LF_{2,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, \tilde{\mu}] - \\
&3 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} a_d^{sr} LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] + \\
&3 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} a_d^{sr} LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - 3 y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} \overline{a_d}^{pr} a_u^{pt} \\
&\quad LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + 3 y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} \overline{a_d}^{pr} a_u^{pt} LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{4} y_e^{i1i2} (\overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} a_d^{sr} + \tilde{\mu}^2 \overline{y_d}^{pr} \overline{y_d}^{st} y_d^{pt} y_d^{sr}) LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}] + \\
&\frac{3}{4} y_e^{i1i2} (-\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} + \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}) LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4} y_e^{i1i2} (\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} - \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}) LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] - \frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \\
&\quad LF_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \frac{3}{4} \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}] + \\
&\frac{3}{8} y_e^{i1i2} (\overline{y_d}^{st} y_d^{pt} \overline{a_u}^{pr} a_u^{sr} - 3 \tilde{\mu}^2 \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr}) LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}] + \\
&\frac{3}{4} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} (3 \overline{a_d}^{pr} a_u^{pt} + \tilde{\mu}^2 \overline{y_d}^{pr} y_u^{pt}) LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] + \\
&\frac{3}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}] - \frac{3}{4} \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_u}^{pr} a_u^{sr} \\
&\quad LF_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \frac{3}{4} \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_u}^{pr} a_u^{sr} LF_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{8} \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_u}^{pr} a_u^{sr} LF_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \\
&\quad LF_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] - \frac{3}{2} \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] + \\
&\frac{3}{2} \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] + \frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} \\
&\quad LF_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{pr} \overline{y_u}^{st} y_u^{pt} y_u^{sr} LF_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \\
&\frac{3}{2} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} (-\overline{a_d}^{pr} a_u^{pt} + \tilde{\mu}^2 \overline{y_d}^{pr} y_u^{pt}) LF_{2,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{3}{2} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} (\overline{a_d}^{pr} a_u^{pt} - \tilde{\mu}^2 \overline{y_d}^{pr} y_u^{pt}) LF_{3,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{2,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{pr} y_d^{sr} y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} LF_{3,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] + \\
&g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] + \frac{1}{2} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] - \\
&2 g_1^2 g_2^2 y_e^{i1i2} LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] + g_1^2 g_2^2 y_e^{i1i2} (-2 m_1 m_2 + \tilde{\mu}^2) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \\
&g_1^2 g_2^2 y_e^{i1i2} LF_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] + \frac{1}{2} g_1^2 g_2^2 y_e^{i1i2} (3 m_1 m_2 - 2 \tilde{\mu}^2) LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \\
&\frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}] + \\
&\frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}] + g_1^2 g_2^2 y_e^{i1i2} LF_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] + \\
&m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] + \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}] - \\
&\frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{3,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}] + \frac{3}{4} g_1^2 g_2^2 y_e^{i1i2} LF_{1,1,1,1,-1}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{2} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{1,1,1,1,0}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{4} g_1^2 g_2^2 y_e^{i1i2} \\
&\quad LF_{2,1,1,1,-2}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{4} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-1}[m_1, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{3}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{3}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad LF_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad LF_{1,1,1,1,0}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&g_1^4 \tilde{\mu}^2 y_e^{i1i2} LF_{2,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{4} g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[m_2, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{4} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-1}[m_2, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{3}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&\frac{3}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] + \\
&\frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_d}^{pr} a_d^{sr} LF_{2,2,1,1,-1}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_d}^{pr} a_d^{sr} LF_{2,2,1,1,-1}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \\
&\frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_2, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{e}}^{r}, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] + \\
&\frac{1}{4} \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \overline{a_e}^{pr} a_e^{sr} LF_{2,2,1,1,-1}[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{4} \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \overline{a_e}^{pr} a_e^{sr} LF_{2,2,1,1,-1}[m_{\tilde{e}}^{t}, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + \\
&\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{p}, m_2, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{1}{2} \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \overline{a_e}^{pr} a_e^{sr} LF_{2,1,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2} \tilde{\mu}^2 \overline{y_e}^{st} y_e^{pt} y_e^{i1i2} \overline{a_e}^{pr} a_e^{sr} LF_{3,1,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{l}}^{s}] + \\
&\frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{r}, \tilde{\mu}] - \\
&\frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{r}, \tilde{\mu}] + \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_d}^{pr} a_d^{sr} LF_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{st} y_d^{pt} y_e^{i1i2} \overline{a_d}^{pr} a_d^{sr} LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] + \\
&3 y_e^{i1i2} \overline{a_d}^{pr} a_u^{pt} (-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}) LF_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&3 y_e^{i1i2} \overline{a_d}^{pr} a_u^{pt} (a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}) LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{4} y_e^{i1i2} (-\tilde{\mu}^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} (a_d^{sr} \overline{a_u}^{st} - 2 \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st})) LF_{2,2,1,1,-1}[\\
&\quad m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{4} y_e^{i1i2} (\tilde{\mu}^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + 3 \overline{a_d}^{pr} a_u^{pt} (-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st})) LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, \\
&\quad m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] + \frac{3}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{4} \tilde{\mu}^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}] - \\
&\frac{3}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] + \frac{3}{4} y_e^{i1i2} \\
&\quad (\tilde{\mu}^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} (a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st})) LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \\
&\frac{9}{8} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{2,1,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} \tilde{\mu}^2 \overline{y_d}^{st} y_e^{i1i2} y_u^{sr} a_d^{pt} \overline{a_u}^{pr} LF_{3,1,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{4} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,2,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \frac{3}{2} y_e^{i1i2} \\
&\quad (\tilde{\mu}^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} (a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st})) LF_{2,1,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \\
&\frac{3}{2} y_e^{i1i2} (\tilde{\mu}^4 \overline{y_d}^{pr} y_d^{sr} \overline{y_u}^{st} y_u^{pt} + \overline{a_d}^{pr} a_u^{pt} (a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st})) LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t}, \\
&\quad m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{2,1,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u}^{st} y_u^{pt} \overline{a_u}^{pr} a_u^{sr} LF_{3,1,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + g_1^2 g_2^2 y_e^{i1i2} \\
&\quad LF_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] + g_1^2 g_2^2 y_e^{i1i2} (m_1 m_2 - \tilde{\mu}^2) LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{e}}^{i2}] - \\
&\frac{1}{4} g_1^2 g_2^2 y_e^{i1i2} LF_{2,1,1,1,-2}[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}] - \frac{1}{4} m_1 m_2 g_1^2 g_2^2 y_e^{i1i2} \\
&\quad LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_2, m_{\tilde{l}}^{i1}] + \frac{7}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \\
&\frac{3}{16} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] + \frac{7}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} \\
&\quad LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] - \frac{3}{16} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}] - \\
&g_1^2 g_2^2 (a_e^{i1i2} (m_1 + m_2) + \tilde{\mu}^2 y_e^{i1i2}) LF_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,1,-1}[m_1, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&m_1 g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} LF_{1,1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, \tilde{\mu}] + \\
&m_1 g_1^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{i1r} a_e^{pi2} LF_{1,1,1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}] - \\
&\frac{1}{2} g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,1,1,1,-1}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}] - \\
&m_2 g_2^2 \tilde{\mu}^2 \overline{y_e}^{pr} y_e^{pi2} a_e^{i1r} LF_{1,1,1,1,1,0}[m_2, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}]\Big)
\end{aligned}
$$